Denuvo

Anti Cheat/ Anti tamper

How it works, How it affects our game, Can it be bypassed.

Presented by:
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Good Morning Everybody, My Name is Luke , Im a first year ethical hacking student with a background in teaching games and id like to welcome you all to my talk about denvo anticheat/anti tamper. Today we will look at what it is, how it work, how it affects our games and can it be bypassed.



So quick show of hands who has heard of denuvo?
So denuvo was created by a company called irdeto
It is currently considered the strongest DRM in gaming and we will get onto how that is changing later.
It is used on a vast variety of games. Mainly triple AAA titles. Most likely you would have encountered it if youve played.
Thisthisthis etc. It is essentially a drm for drms. It protects other drms that give you licneses for your games.

SOUNDS PRICEY?

Well YES

- We don't really know how much denuvo costs. It seems to vary per scenario as well as length of use.
- But Based on the Crytek hack leaked documents reveal for crysis remastered:
  - €140,000 For 12 months
  - €2000 every month after
  - Plus fees for influx of users and additional storefronts

WOW big titles is what your thinking so it must cost a pretty penny. Well yes, we actually dont have much in the ways of pricing with denuvo it seems to change depended on the game or the use case how much work it will take to implement that sort of thing. The one piece of information we have comes from a hack of crytek back in 2020. Baded on the leaked documents we have seen that the fee was 140,000 euros for 12 months + 2000 extra month after. As well as this there where fees such as
€60 000 extra fee for products that receive over 500 000 unique activations in 30 days;

€10 000 extra fee for each storefront (digital distribution service) the product gets put on.

WHY?

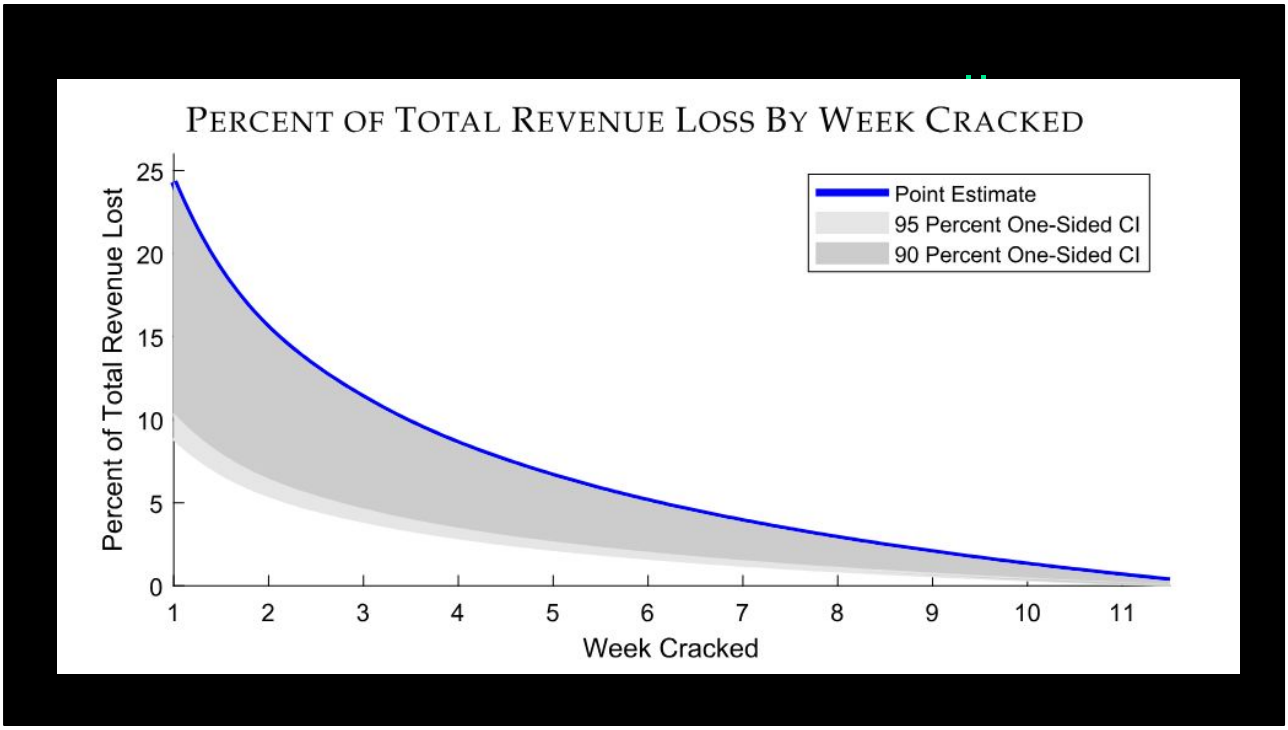
SO WHY IS IT STILL USED?

In 2016 : 90% of gamers had pirated a game, 25% of Gamers had pirated 50 or more games

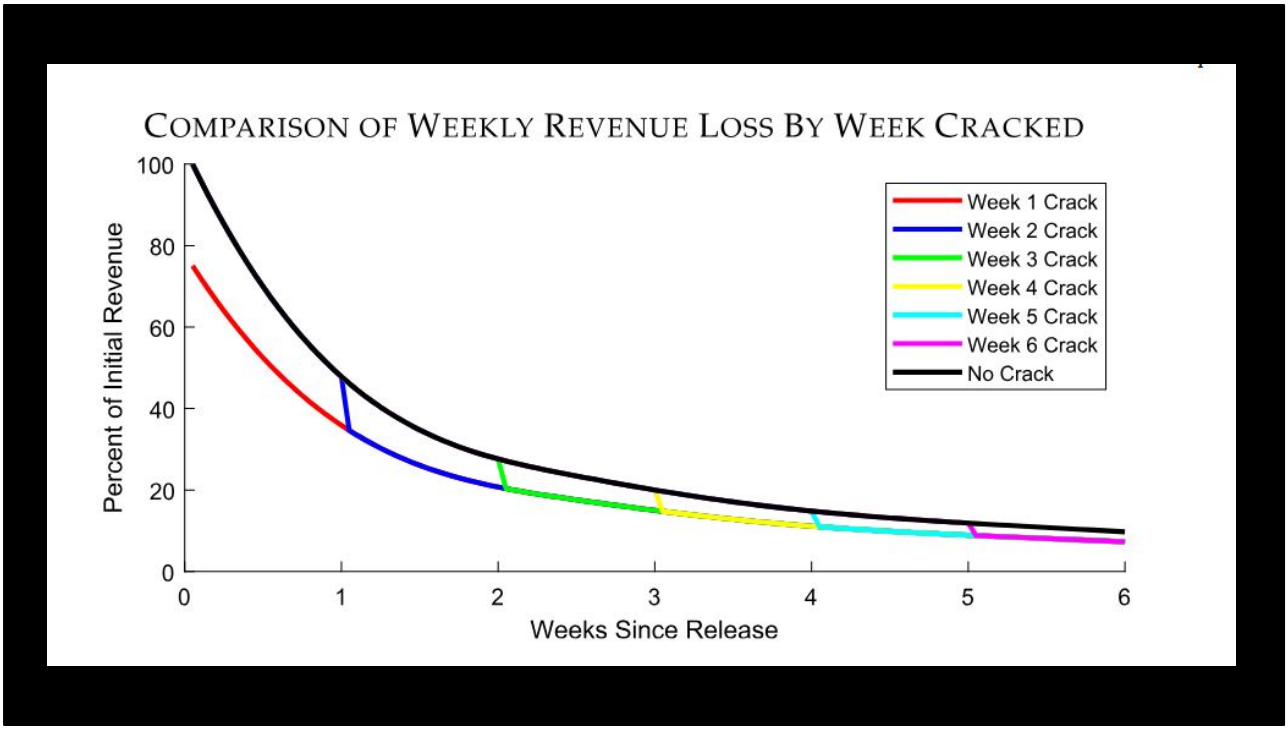
Irdeto claim that they have a 180 day average crack protection record

https://www.pcgamer.com/pc-piracy-survey-results-35-percent-of-pc-gamers-pirate/
https://irdeto.com/hubfs/resources/data-sheets/denuvo-anti-piracy-one-pager.pdf

According to a survey done of over 50000 people. by PC gamer in 2016 it was recorded that 90 percent of gamers had pirated a game before and 25% had pirated 50 or more games. Now we can obviously take this with a pinch of salt as it was not a controlled survey or study. But it still shows that piracy is more than relevant and could affect a games income by a hefty amount. Irdeto say that they can protect a game for an average of 180 days before being cracked. So does this justify the cost. Does it protect a games income enough.

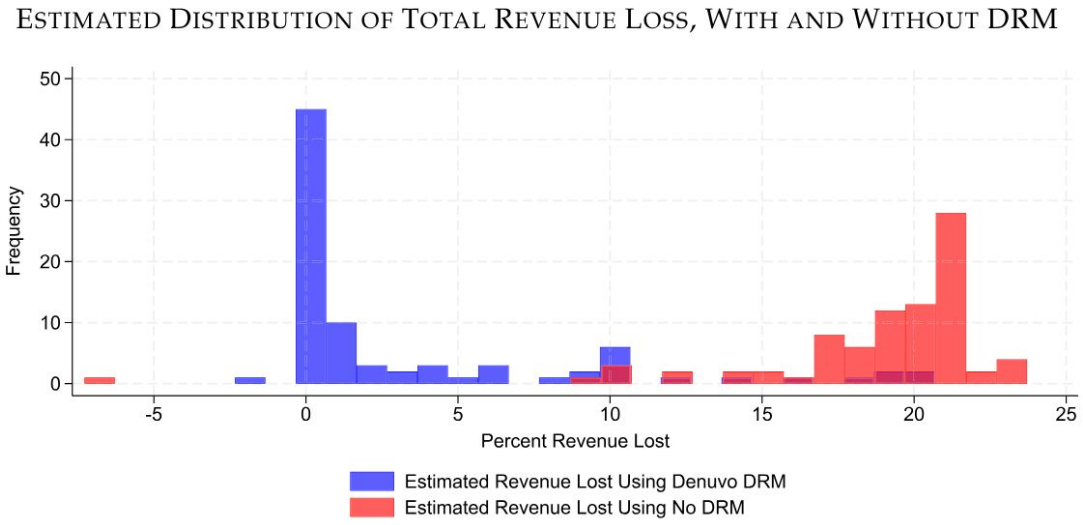


So lets have a look here we have a graph where we can see that a denuvo crack can mean up to 25 percent of a games revenue is lost if the game is cracked early on where theres a lot of excitement around the release and the price is still at it max. And then obviously this goes down with time



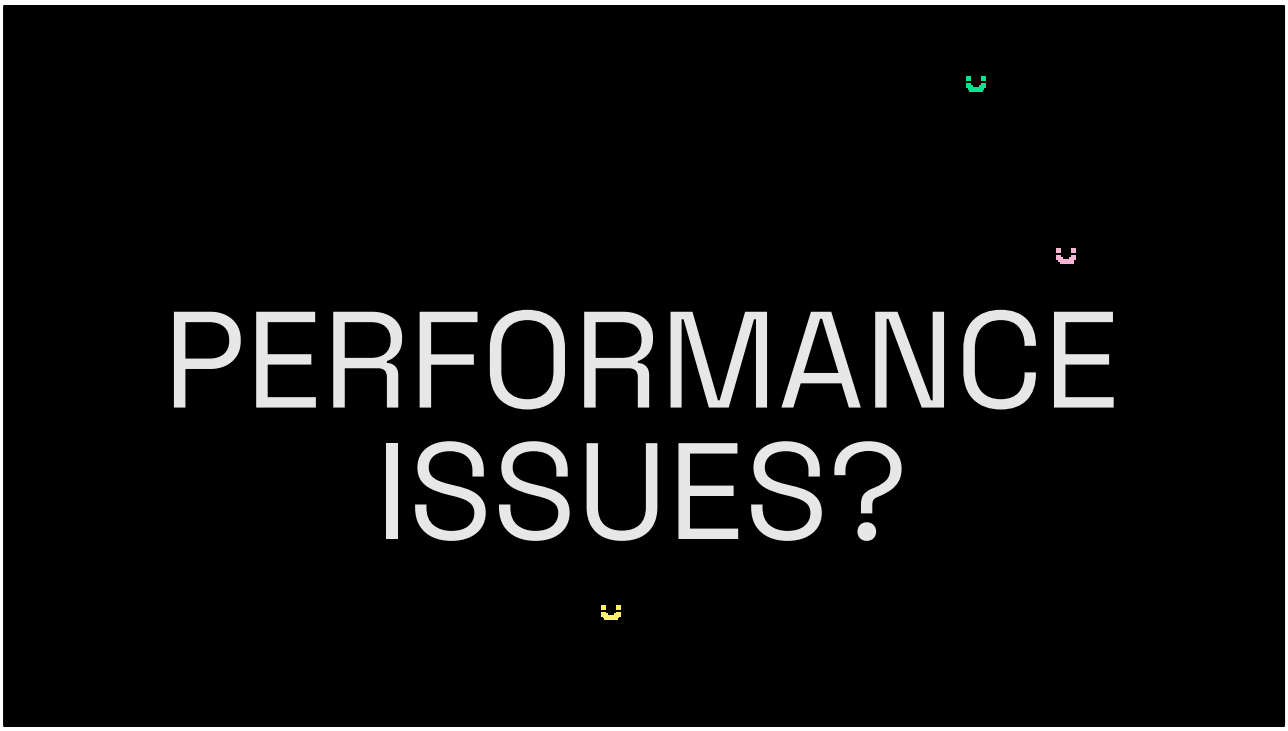
If we compare these figures to an average revenue loss of a uncracked game. We can still see that thre is a significant impact on revenues that we can see levels out after around 6 weeks. Which means if irdeto is correct in their claim. Denuvo will be able to effectively protect a games revenue for most of its relevant life span.

A Study from William M. Volckmann II



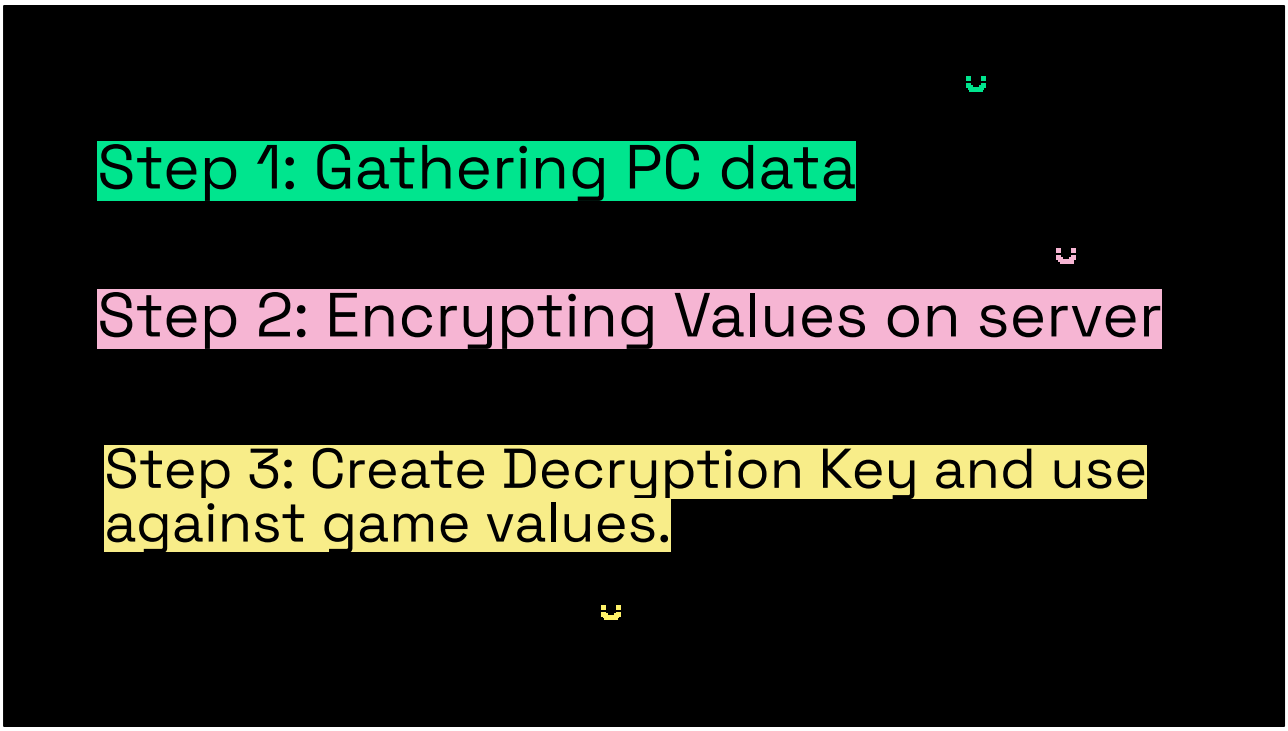
https://www.sciencedirect.com/science/article/abs/pii/S1875952124002532?fr=RR-2&ref=pdf_download&rr=9b239dffc8f44190

This is reinformed by this graph. Where we see that denvo protected games lose a lot less revenue than non drm protected games. By a significant figure. All of these graphs where taken from a study from William Volckmann titled Revenue effects of Denuvo digital rights management on PC video games.

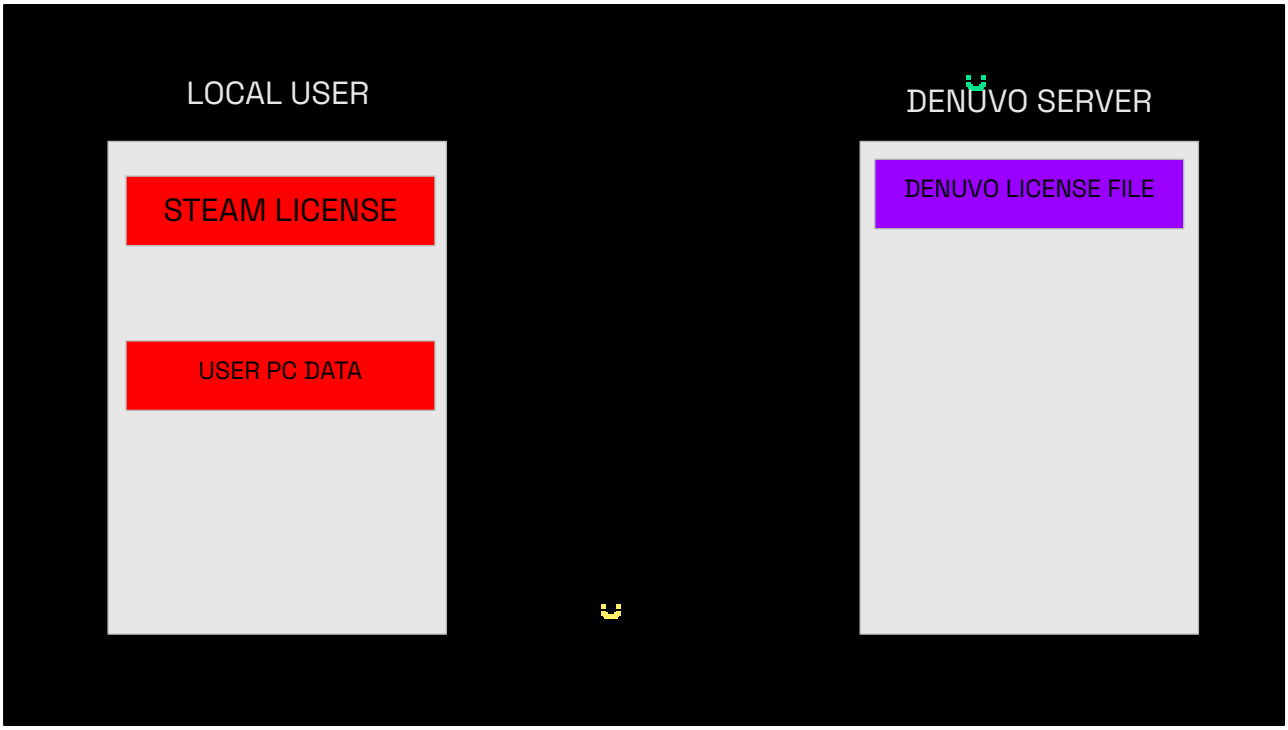


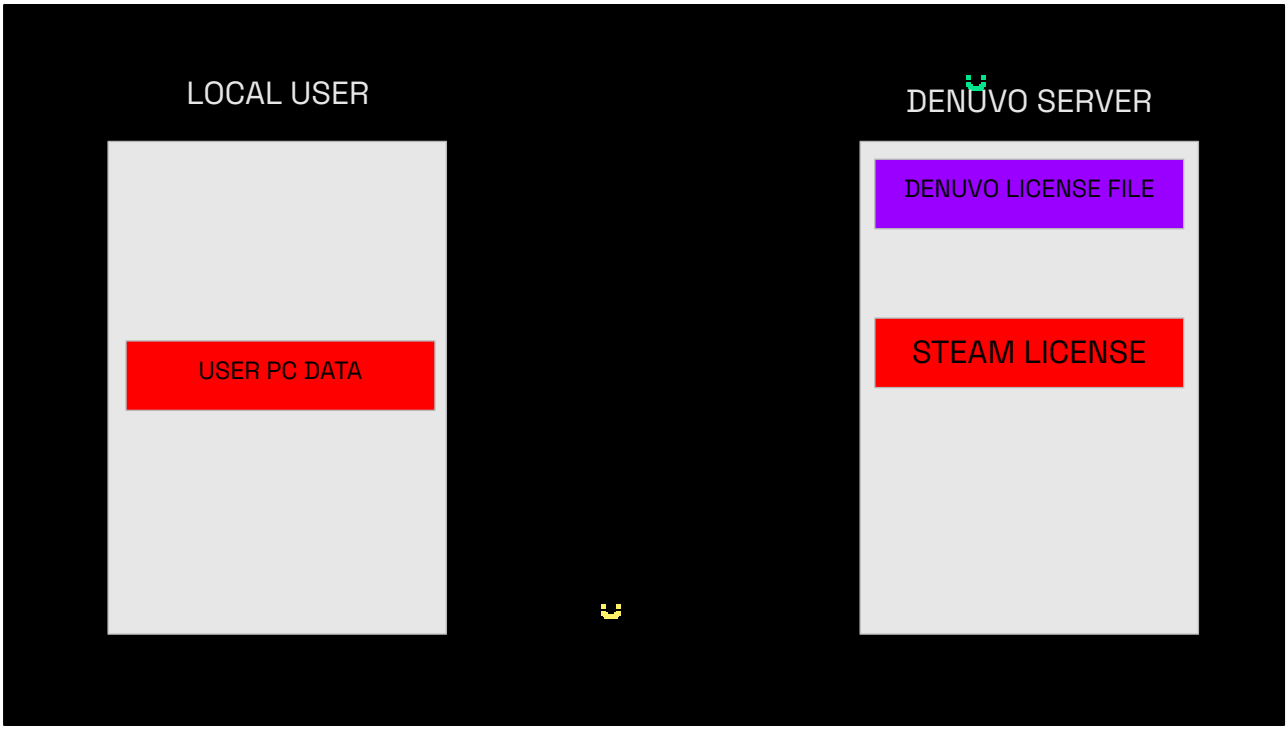
There are loads of anecdotes telling us that denuvo ruins the performance of our games. However there seems to be no concrete evidence. The games in which there where massive increases to performance after a crack is either when there has been another 3rd pary drm involved or the cracker tweaked settings to boost performance. There are not many test comparisons online but the ones that I did find conclude that there is no evidence that denuvo affects performance. Theoretically performance could be affected during runtime checks however tihs is not an issue if checks occur in places such as loading screens,

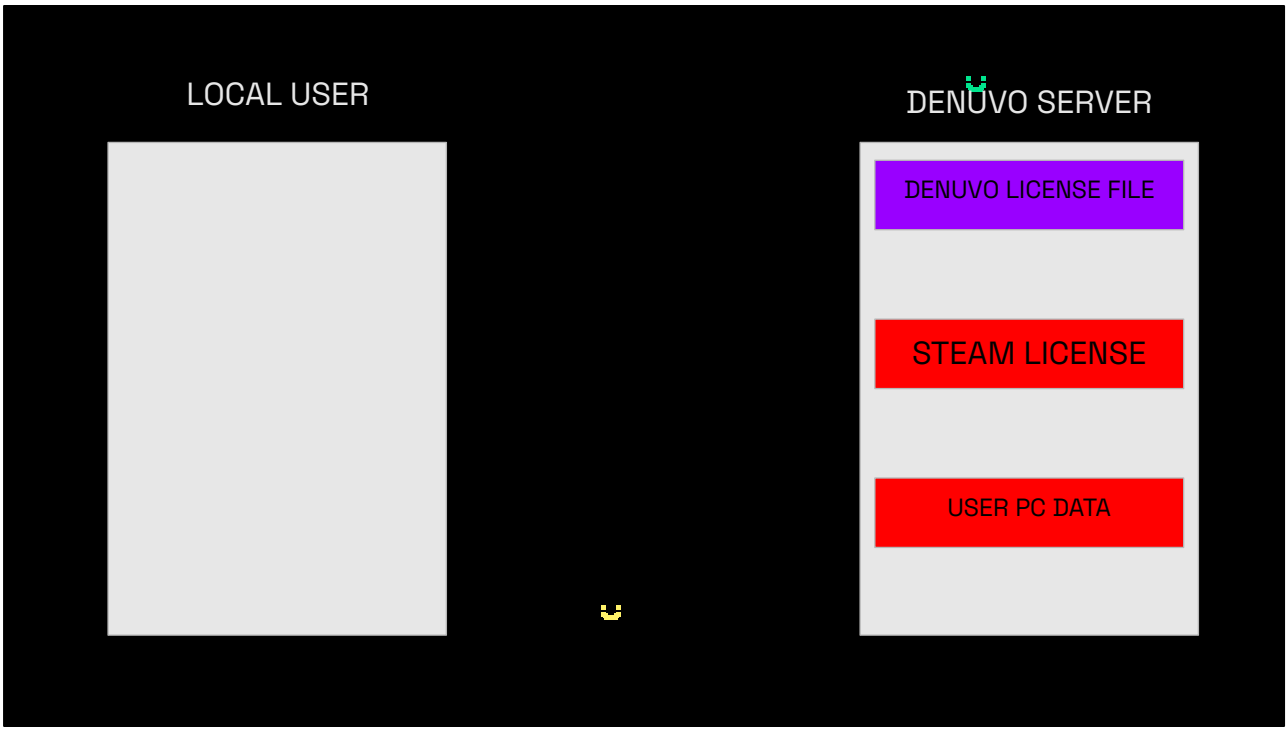
SO HOW DOES IT ALL WORK THEN?

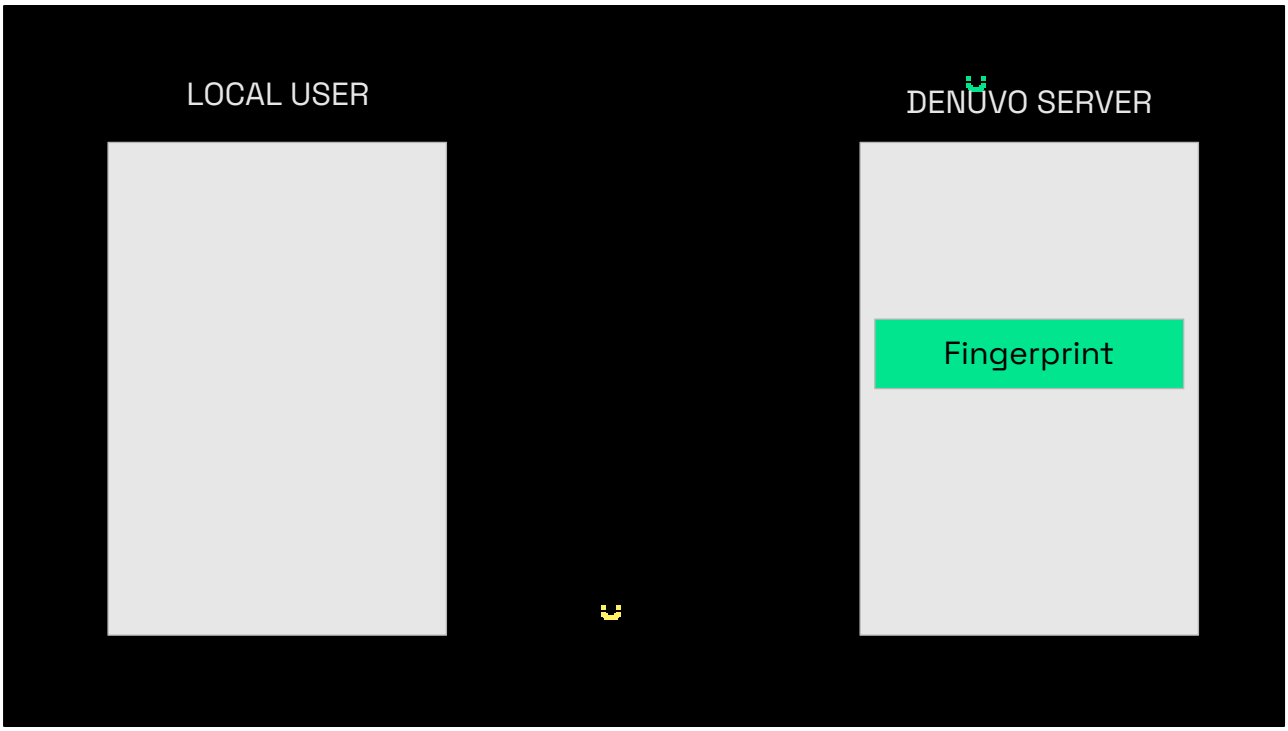


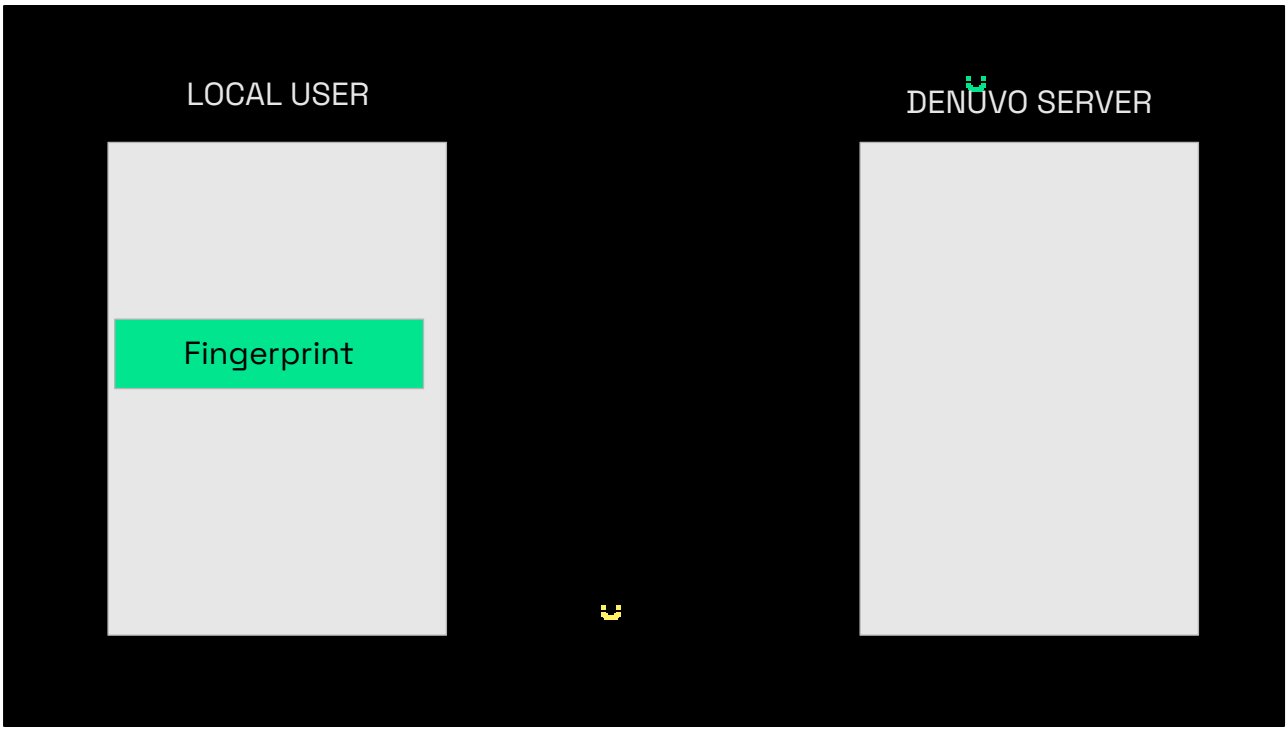
So how does it all work then. So on boot denuvo will start off by getting pc data of the user as well as other DRM data for example steam games can have steam tickets as proof of ownership that can also be collected. This is then sent to the denuvo server where magic happens and essentially a fingerprint is created. This is the liscence key that can then be checked by the denvuo anti tamper. Every time the game is booted denuvo will check to see if the pc data is the same. If it is not the user will need to go back online to create a new liscence file

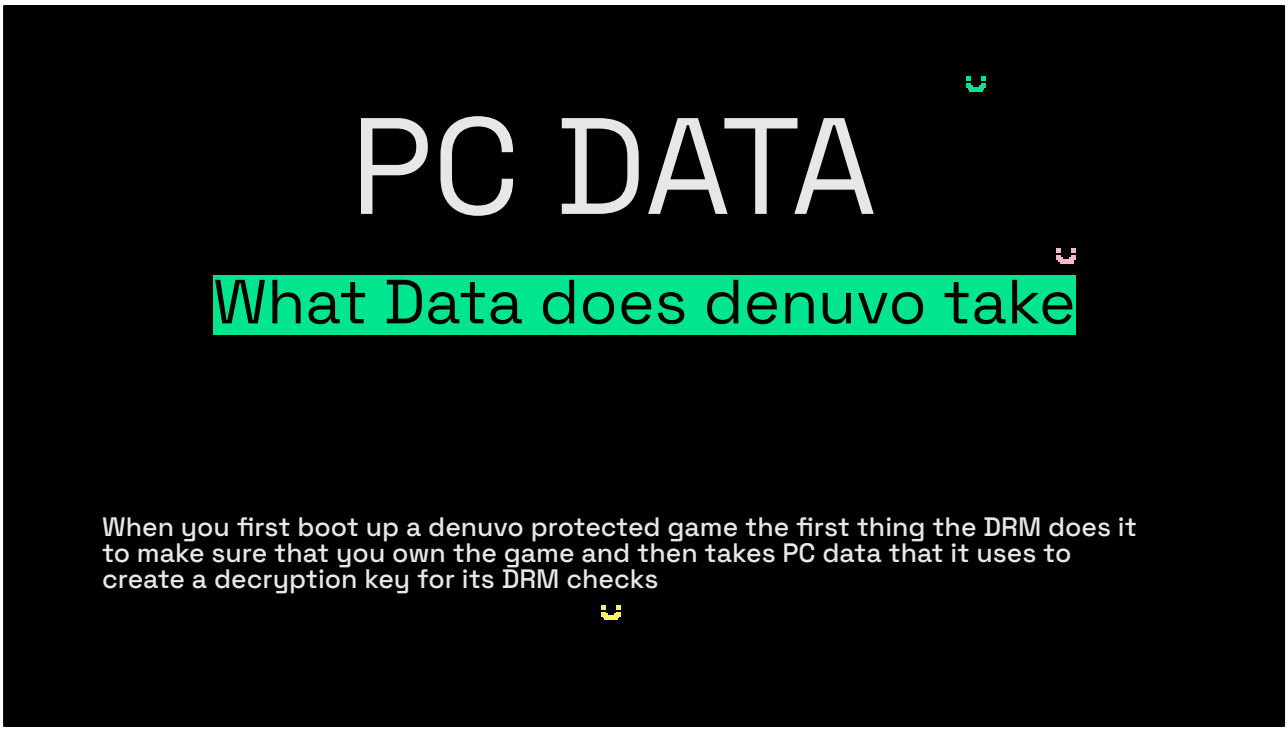

LOCAL USER
STEAM LICENSE
USER PC DATA
DENUVO SERVER
DENUVO LICENSE FILE


LOCAL USER
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STEAM LICENSE


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DENUVO LICENSE FILE
STEAM LICENSE
USER PC DATA


LOCAL USER
DENUVO SERVER
Fingerprint


LOCAL USER
DENUVO SERVER
Fingerprint



So what data does denuvo actually gather. Well this can be generic pc data such as the hardware information. It can include things like software versions, cpu architecture, and even things such as storage drive information, windows information, directory paths etc. Basically a lot of information about your pc to make the key

Tamper Checks

How does denuvo use this key?

So essentially when a game is made. Certain functions will be selected to with a known outcome. Certain parts of the machine code will then be removed from the instructions and these parts will be the data that will then need to be decrypted by the key. These known functions are mathematically combined with the hardware info in collected by denuvo creating a license file. This is then run in a virtual machine to check to see whether the instructions add up. If they do not then the game will crash. This is done on startup but also during runtime for a lot of games.

CRACKING DENUVO

So Can it be Bypassed

Hint: YES

A BIT OF HISTORY

When we talk about cracking denuvo we need to look at time periods.

There are multiple denuvo version, and multiple ways to crack them

Denuvo has been cracked many a times before.

Denuvo is constantly evolving so gets harder a[illegible]r.

EMKDEVS

For the older versions of Denuvo there where a few people/groups that could crack denuvo

So to start off with a bit of history we need to look at who was the first group to crack denuvo. This goes to a chinese group called 3dm. And the game being dragon age inquisition in 2014. After that time went by and denuvo evolved but there where a few main groups that were able to crack denuvo.

CPY: cpy are responsible for many cracks from denuvo, they are known to be responsible for cracking the first of denuvo version 3 they where efficient in cracking games very quickly after release. Some notable cracks RE7, Metro Exodos, death stranding,crysis remastered, just cause 4. They went off the scene around 2020 and have not been seen since

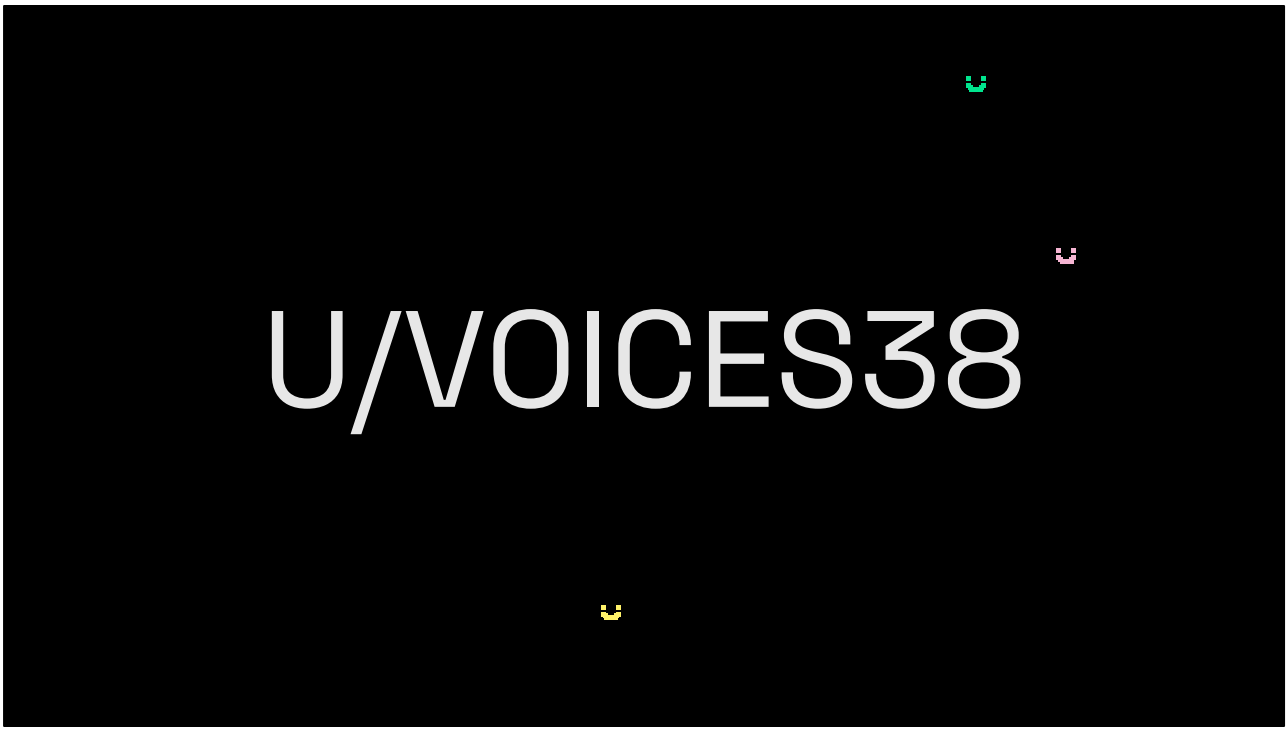
CODEX: Codex where also responsible for a fair few cracks such as re2, far cry 5 and team sonic racing. They Where one of the first groups responsible for cracking denuvo but retired in 2022 citing that they where created as competition for another group but now there was not enough competition.

Empress: Empress is probably the most interesting and renowned denuvo cracker. Thought to be one individual and not a group they are most notable known for games such as hogwarts legacy, Far Cry 6, RE4 Remake,rdr2. I could honestly talk about empress for the whole half an hour. To cut the story short they where a little crazy, unlike the groups who crack popular games and then call for donations. You could pay empress 500 dollars to crack a game. They also had a lot of controversial views some of which where transphobic which lead to a falling out with one of the biggest repackers. Empress made many wild claims over the time and often got in a lot of arguments. They made the claim that denuvo once offered them 200k a month to work for them which we will take with a big pinch of salt. Their last crack was around 2023 but they officially made their retirement statement a couple of months ago.

MKDEV: MKDEV has a lot of relevance later on. They are most known for retiring, coming out of retirement to crack football manager or fifa and then retiring again. They are actually a talented group who as I said have a lot of relevance now. Up until a couple of months ago they had not been seen since 2023

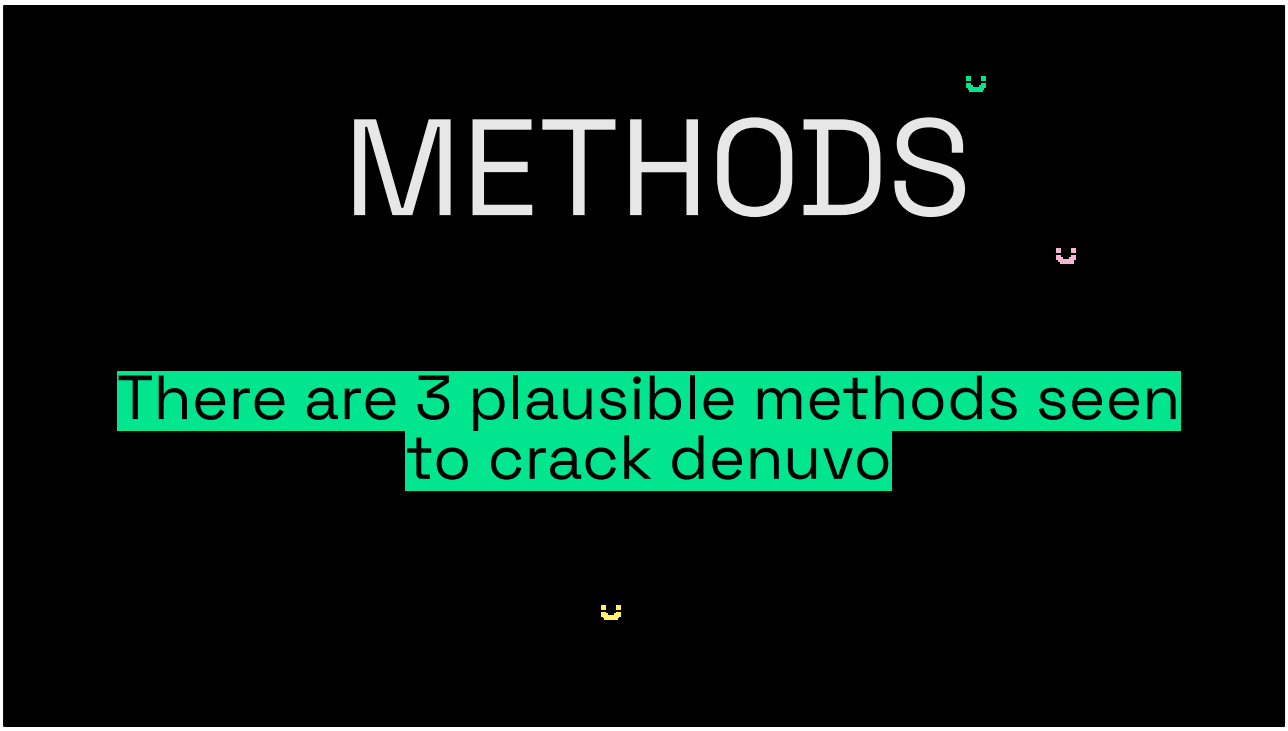
SO EVERYONE'S RETIRED?

This is where it gets interesting?



So Denuvo cracking has been rather dry in the last couple of years. There have still been some people such as RUNE cracking older games but they haven't cracked Denuvo for at least a year. When to come to modern denuvo though there has really been not much movement. Until around 3 months ago where a new player came onto the scene. Voices38. Voices38 has been cleaning up the denvuo uncracked games lists and has been regularly cracking denuvo games. Including some more modern title such as the deas space remake, hifi rush and star wars squadron. They have recently confirmed that they are working on tools for denuvo 2024

Voices 38 has been cleaning up the uncracked games list and is developing tools for use of the latest denuvo games. At this point we are pretty caught up until 2020-2021 ish. There has been a pause on voices 38 cracks but speculation is he is working on tools for newer versions of denuvo.

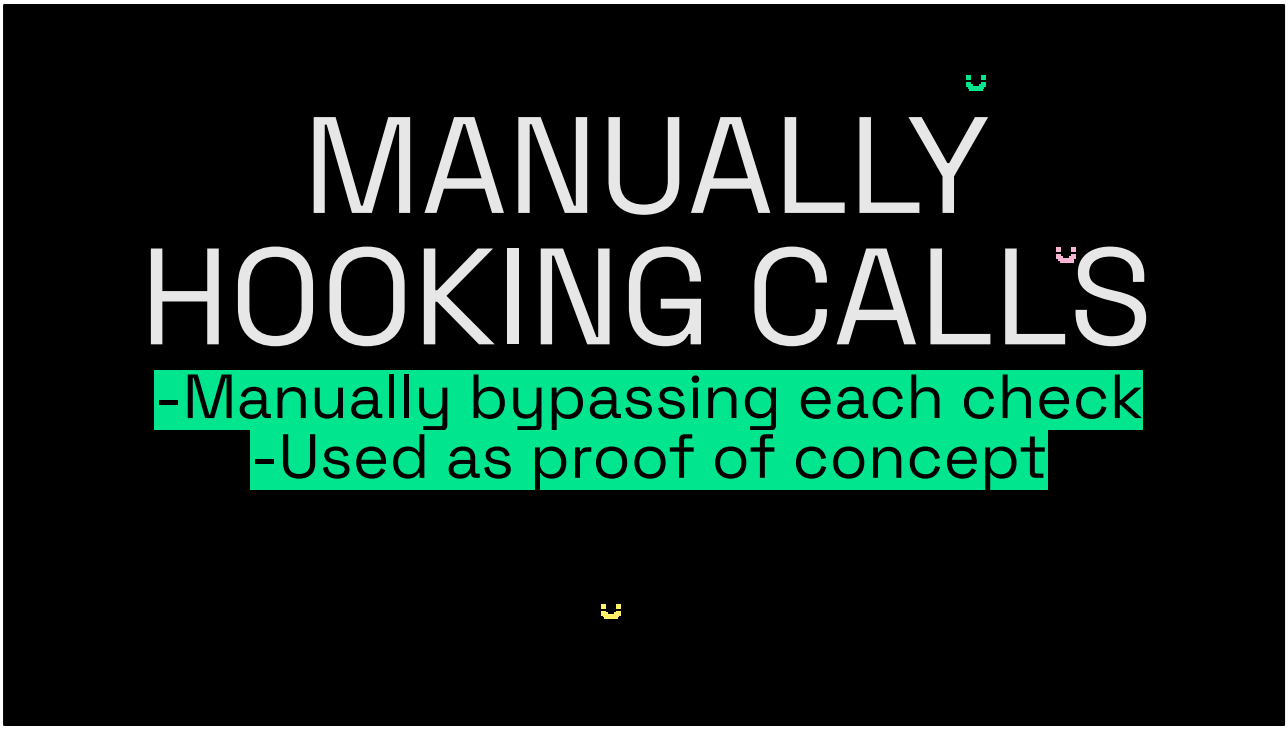


There are 3 plausible methods to cracking denuvo.

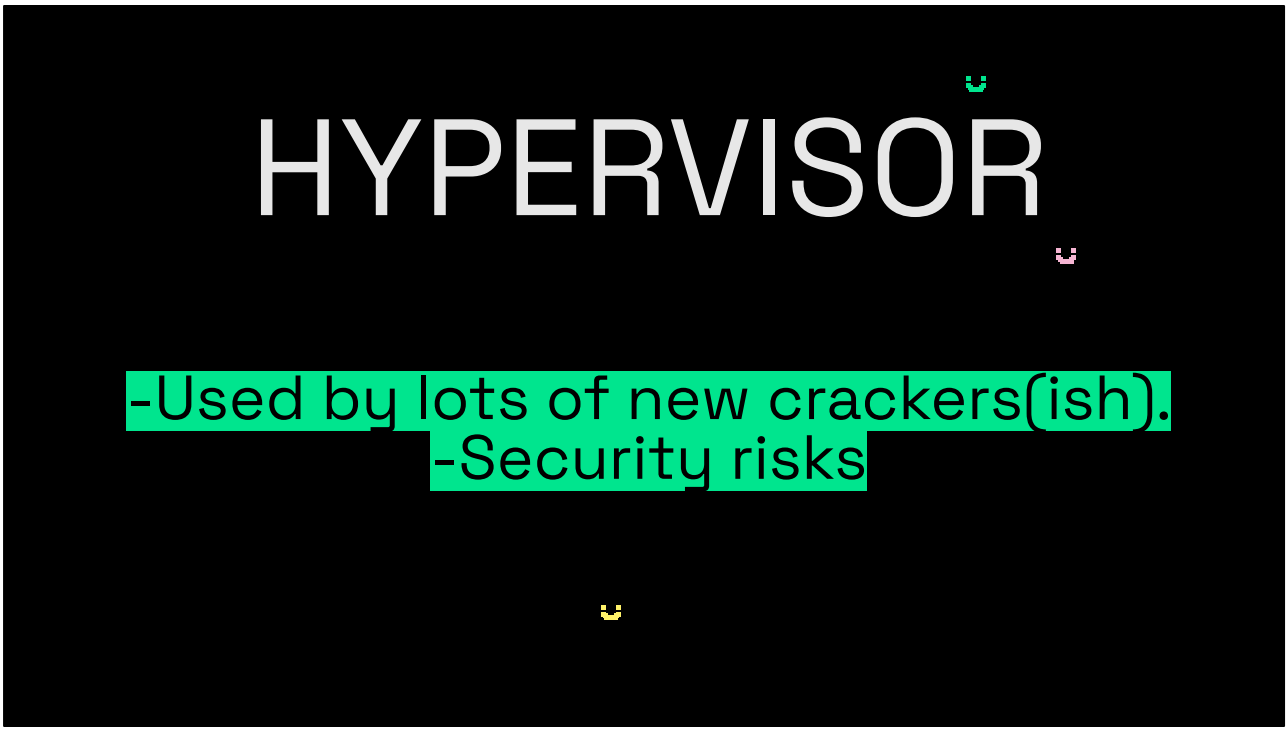
LICENSE HASHES

Used by the likes of cpy and voices38

This is where the cracker bypasses the denuvo checks by satisfying the license requirements without the need for said liscence, this means that any game can work on any pc. The method voices38 uses is to replace a .dll file call with his own which then essentially runs his keygen which satisfies the license requirements.



This is more of a proof of concept way to crack denuvo in which you can look for each system call and check and manually hook it onto new information of your choice



This is probably the most straightforward way. This is essentially running the game in a virtual machine that aligns with the correct hardware requirements. This has been used for a while but recently MKDEV cam back onto the scene to retire again leaving information and methods on hot to create a hypervisor for denuvo. This has made bypassing denuvo accessible for a lot more people. The major downside to this is that it is usually cpu specific and also poses massive security risks as you have to essentially turn off all security features on your computer and run a kernel level driver.

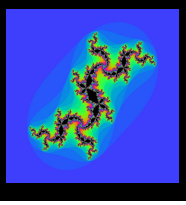
THE FUTURE OF DENUVO

My thoughts?

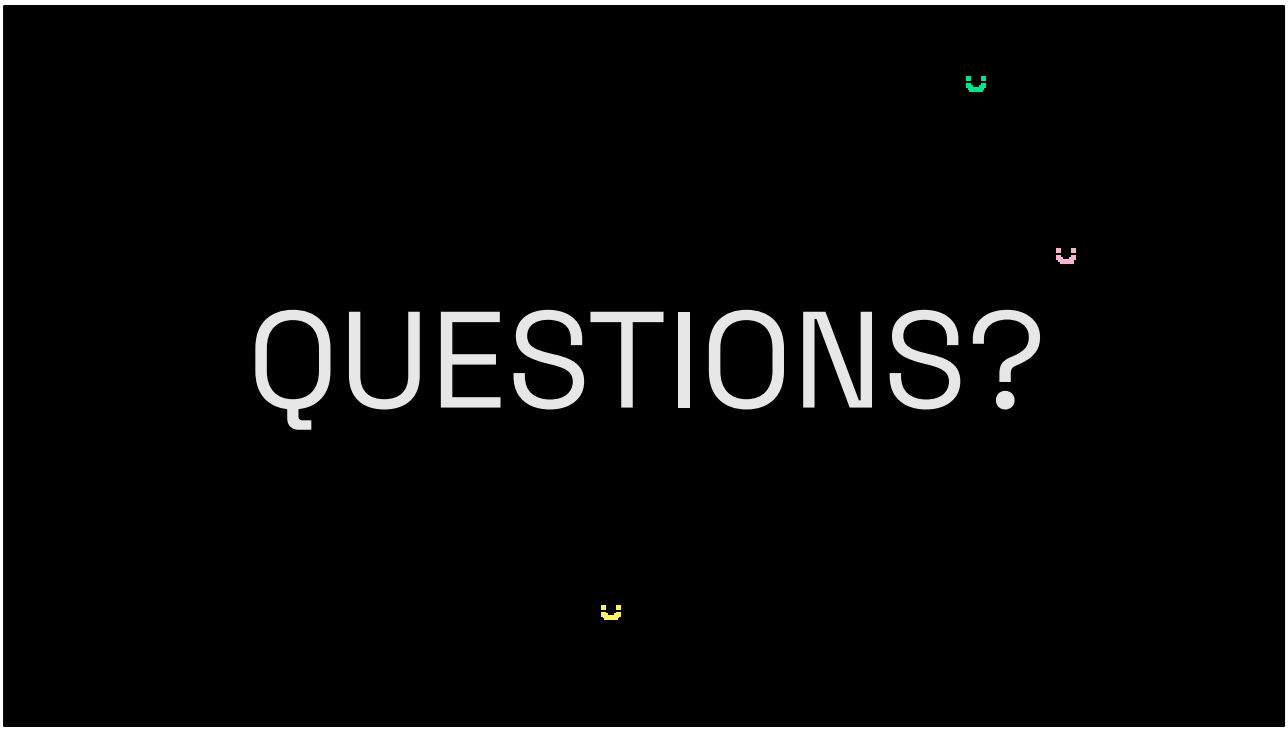
Further Reading

https://momo5502.com/posts/2025-10-03-reverse-engineering-denuvo-in-hogwarts-legacy/

https://connorjaydunn.github.io/blog/posts/denuvo-analysis/






QUESTIONS?